VALORES DE LONGITUD ASIGNADOS

$l_1 = 10$ $\quad \theta_1 = 30°$ $\quad \theta_3 = 30°$

$l_2 = 10$ $\quad \theta_0 = 30°$ $\quad \theta_4 = -30°$

$l_3 = 5$ $\quad \theta_2 = 30°$

ENFOQUE GEOMÉTRICO

$\theta_2 = 30°$

$\theta_2 + \theta_3 = 30 + 30 = 60°$

$\theta_2 + \theta_3 + \theta_4 = 30 + 30 - 30 = 30$

PROYECCIÓN EN EL PLANO VERTICAL

$$r = l_1 \cos(\theta_2) + l_2 \cos(\theta_2 + \theta_3) + l_3 \cos(\theta_2 + \theta_3 + \theta_4)$$

$$r = 10 \cos 30 + 10 \cos 60 + 5 \cos 30$$

$$r = 10\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)(10) + 5\left(\frac{\sqrt{3}}{2}\right) = 5 + \frac{15\sqrt{3}}{2} = 18..$$

$$z = 10 \sin(30) + 10 \sin(60) + 5 \sin(30)$$

$$z = 10(1/2) + \frac{10\sqrt{3}}{2} + 5\left(\frac{1}{2}\right) = \frac{15}{2} + 5\sqrt{3} = 16.160$$

COORDENADAS TRIDIMENSIONALES

$$x = r \cos(\theta_1) = 17.99 \cos 30 = 17.99\left(\frac{\sqrt{3}}{2}\right) = 15.580$$

$$y = r \sin(\theta_1) = 17.99 \cdot \sin(30) = 17.8\left(\frac{1}{2}\right) = 8.995$$

$$Z = 16.160$$

VECTOR DE POSICIÓN FINAL

$$P = \begin{bmatrix} 15.580 \\ 8.995 \\ 16.166 \end{bmatrix}$$

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# Matrices individuales

$$A_1 = \begin{bmatrix} \cos(30) & 0 & \sin 30 & 0 \\ \sin(30) & 0 & -\cos 30 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & -\sqrt{3}/2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \cos 30 & -\sin 30 & 0 & 10\cos 30 \\ \sin 30 & \cos 30 & 0 & 10\sin 30 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 & 5\sqrt{3} \\ 1/2 & \sqrt{3}/2 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} \cos 30 & -\sin 30 & 0 & 10\cos 30 \\ \sin 30 & \cos 30 & 0 & 10\sin 30 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 & 5\sqrt{3} \\ 1/2 & \sqrt{3}/2 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} \cos(-30) & -\sin(-30) & 0 & 5\cos(-30) \\ \sin(-30) & \cos(-30) & 0 & 5\sin(-30) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

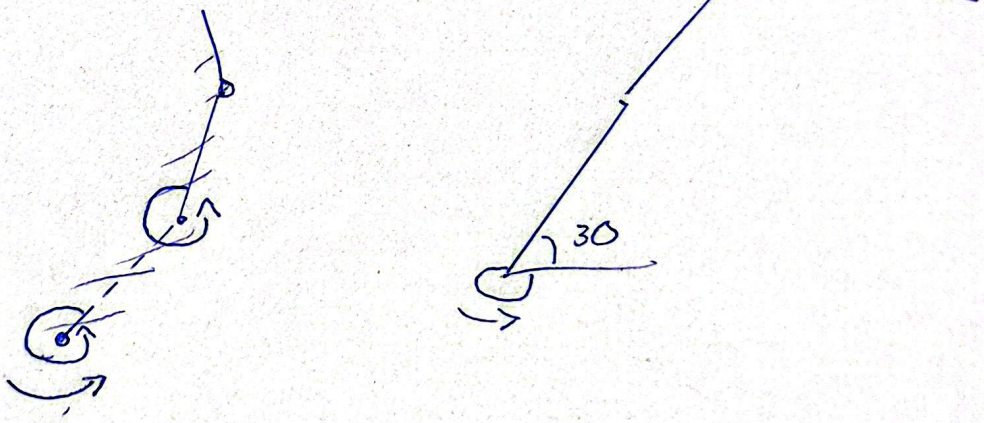
$$\begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 & 5\sqrt{3}/2 \\ -1/2 & \sqrt{3}/2 & 0 & -5/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$T_0^4 = A_{1-4} = \begin{bmatrix} 3/4 & -\sqrt{3}/4 & 1/2 & (45/4 + 5\sqrt{3}/2) \\ \sqrt{3}/4 & -1/4 & -\sqrt{3}/2 & (5/2 + 15\sqrt{3}/4) \\ 1/2 & \sqrt{3}/2 & 0 & (15/2 + 9\sqrt{3}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_0^4 = \begin{bmatrix} 0.750 & -0.433 & 0.5 & 15.58 \\ 0.433 & -0.250 & -0.866 & 8.995 \\ 0.5 & 0.866 & 0 & 16.160 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 15.580 \\ 8.995 \\ 16.160 \end{bmatrix} \rightsquigarrow \text{Posición}$$




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